Subnetting Design Strategy
Kestrel Industries

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Introduction

This document breaks down how I've structured the IPv4 network for Kestrel Industries. Starting with the base block of `10.20.5.0/24`, the goal was to carve out enough space for every department and WAN link without being wasteful. I used Variable Length Subnet Masking (VLSM) to make sure each group gets exactly what they need, no more, no less, while keeping things organized for future growth.

The Subnetting Table

I've organized the subnets starting from the largest requirements down to the smallest. This is the most efficient way to pack the addresses and avoid overlaps.

Network Name	Hosts Needed	Network Address	Usable Address Range	Broadcast Address	Subnet Mask
Customer A	28	10.20.5.0	10.20.5.1 – 10.20.5.30	10.20.5.31	255.255.255.224
HR	14	10.20.5.32	10.20.5.33 – 10.20.5.46	10.20.5.47	255.255.255.240
Procurement	12	10.20.5.48	10.20.5.49 – 10.20.5.62	10.20.5.63	255.255.255.240
Engineering	11	10.20.5.64	10.20.5.65 – 10.20.5.78	10.20.5.79	255.255.255.240
Customer B	10	10.20.5.80	10.20.5.81 – 10.20.5.94	10.20.5.95	255.255.255.240

IT	8	10.20.5.96	10.20.5.97 – 10.20.5.110	10.20.5.111	255.255.255.240
Finance	6	10.20.5.112	10.20.5.113 – 10.20.5.118	10.20.5.119	255.255.255.248
Executive	6	10.20.5.120	10.20.5.121 – 10.20.5.126	10.20.5.127	255.255.255.248
Sales	4	10.20.5.128	10.20.5.129 – 10.20.5.134	10.20.5.135	255.255.255.248
R&D	4	10.20.5.136	10.20.5.137 – 10.20.5.142	10.20.5.143	255.255.255.248
Admin-Core (WAN)	2	10.20.5.144	10.20.5.145 – 10.20.5.146	10.20.5.147	255.255.255.252
Ops-Core (WAN)	2	10.20.5.148	10.20.5.149 – 10.20.5.150	10.20.5.151	255.255.255.252
Core-West (WAN)	2	10.20.5.152	10.20.5.153 – 10.20.5.154	10.20.5.155	255.255.255.252
Core-East (WAN)	2	10.20.5.156	10.20.5.157 – 10.20.5.158	10.20.5.159	255.255.255.252
West-East (WAN)	2	10.20.5.160	10.20.5.161 – 10.20.5.162	10.20.5.163	255.255.255.252

Design Logic

I didn't just pick these numbers out of a hat. For the larger groups like `Customer A` (28 hosts), I used a `/27` mask (255.255.255.224), which gives us 30 usable spots perfect for their needs. For the tiny point-to-point links between routers (the WANs), a `/30` mask (255.255.255.252) was the way to go since we only need 2 IPs for those.

Routing Plan

To keep the traffic moving, I've set up two main protocols:

RIP v2: This is running on the `Admin`, `Core`, and `Ops` routers. It's simple, effective for these segments, and handles our VLSM setup without any issues.

OSPF Area 1586: (Using my student number 15186: Process 15, Area 86). This is the heavy lifter for the `Core`, `West`, and `East` routers. It's fast and scales much better for the main backbone of the network.

Final Thoughts

This setup is clean, logical, and follows the brief to the letter. Every host has a home, and the routing is set up to ensure that everyone can talk to everyone else (and reach the web server) without any hiccups.

End of Report